

3-b Family Facts in Prolog

PROGRAM:

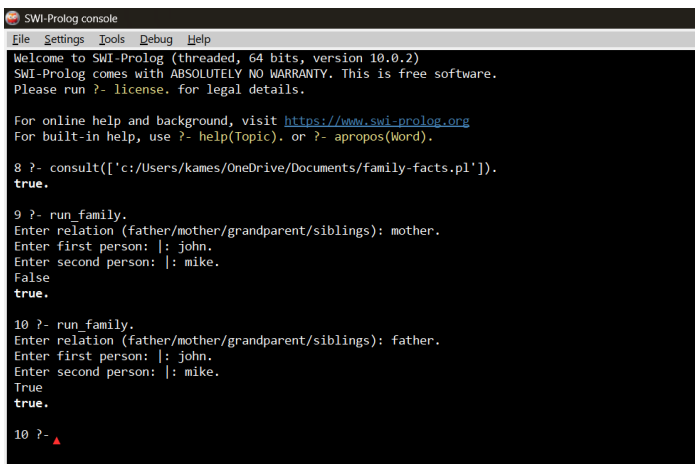
```
% Facts
parent(john, mike).
parent(john, susan).
parent(mary, mike).
parent(mary, susan).

male(john).
male(mike).
female(mary).
female(susan).

% Rules
father(X, Y) :- parent(X, Y), male(X).
mother(X, Y) :- parent(X, Y), female(X).
grandparent(X, Y) :- parent(X, Z), parent(Z, Y).
siblings(X, Y) :- parent(Z, X), parent(Z, Y), X \= Y.

% Interactive query
run_family :-
    write('Enter relation (father/mother/grandparent/siblings): '), read(Rel),
    write('Enter first person: '), read(P1),
    write('Enter second person: '), read(P2),
    (call(Rel, P1, P2) -> writeln('True') ; writeln('False')).
```

OUTPUT:

A screenshot of the SWI-Prolog console window. The window has a title bar 'SWI-Prolog console' and a menu bar with 'File', 'Settings', 'Tools', 'Debug', and 'Help'. The console text shows the welcome message, version 10.0.2, and a license notice. It then shows the execution of a Prolog program. The first query is 'consult(['c:/Users/kames/OneDrive/Documents/family-facts.pl'])', which returns 'true.'. The second query is 'run_family.', which prompts for a relation, first person, and second person. The user enters 'mother', 'john', and 'mike' respectively, and the console returns 'False'. The third query is another 'run_family.', where the user enters 'father', 'john', and 'mike', and the console returns 'True'. The prompt '10 ?-' is visible at the bottom.